

# Informe Café → Azul — copia fiel (sin suavizar)

Generado: 2026-07-31 11:07 UTC

PDF: INFORME\_COPIA\_AZUL\_FIEL\_V22.pdf

Repasada general (evidencia = 16 parejas AAC / 32 M4A):

- Separación nivel vs timbre: gain  $\approx -12$  dB (consenso V12/V15/V17/V4.1)
- EQ operativa: V22.0 operativa · air v22\_hard\_10k
- pipeline\_version: V22.0-operative
- source\_variant: v22\_winsor\_fretted\_tight\_floor+v22\_hard\_10k
- air\_policy: v22\_hard\_10k
- Gain operativa: -12.939 dB
- Hold-out crítico 0.5–8 kHz: 4.333 dB
- Sin suavizado regional ni contracción EQ×reliability
- Aire 10–18 kHz: taper V20 (eq\_copy  $\neq$  V19 cruda encima de 10 kHz)

Qué hacemos mejor que V4.1: DPSS, F0 refine, matching acústico, offsets estimados, repro bit-idéntica, métrica render Café+EQ vs Azul.

Qué adoptamos de V4.1 para fidelidad: pesos SNR/ciclos/AAC, pair-first, confianza de alineación, repetibilidad de fase, neutralidad energética.

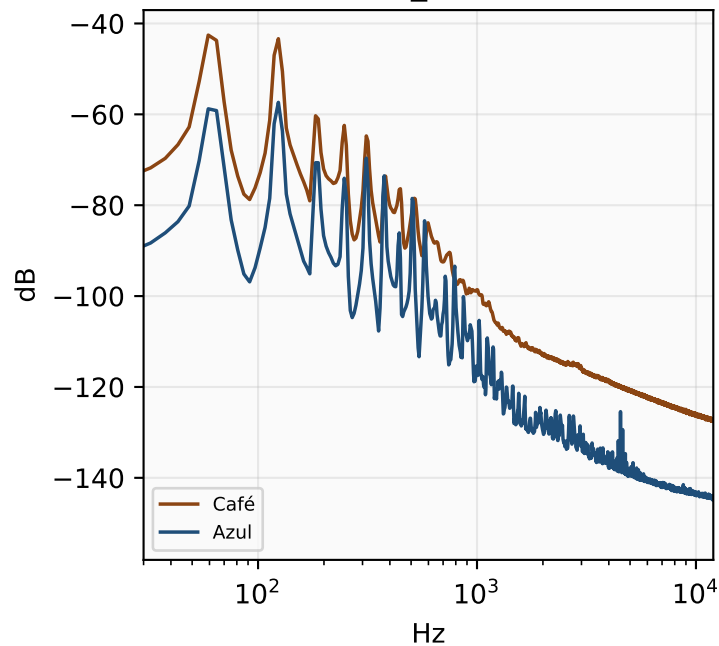
Qué NO adoptamos como EQ principal: smooth held-out, EQ×fiabilidad, multiplicadores físicos fijos 0.62/0.55/0.85.

Contenido:

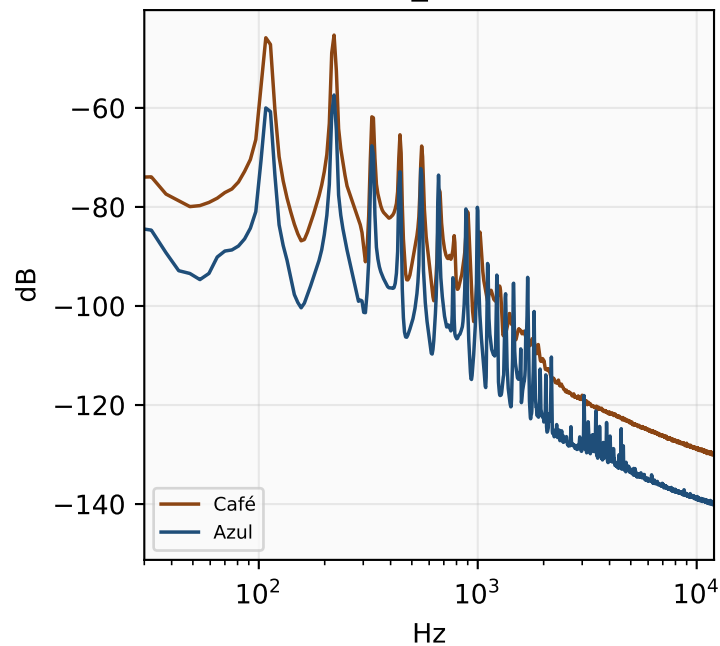
- 1) Espectros Café vs Azul (pares representativos)

# 1. Espectros Café vs Azul (Welch, frames activos)

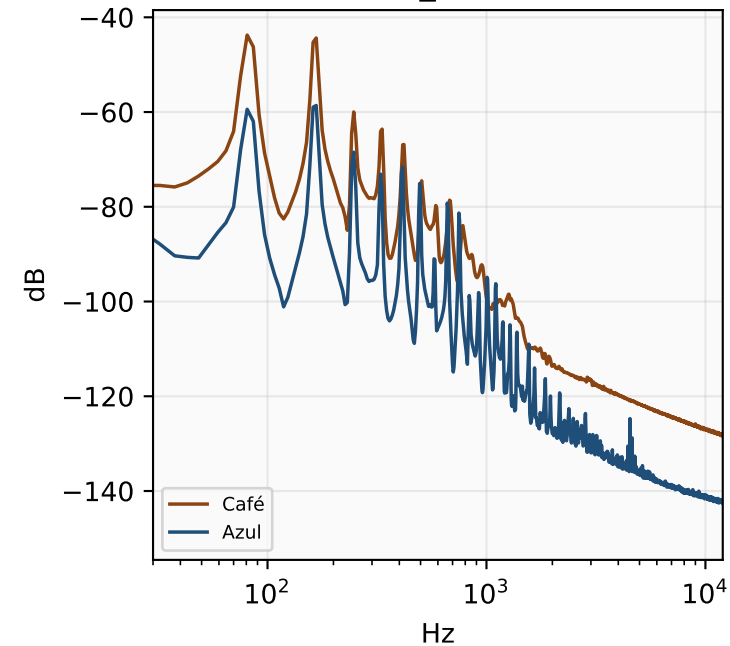
B<sub>12</sub>



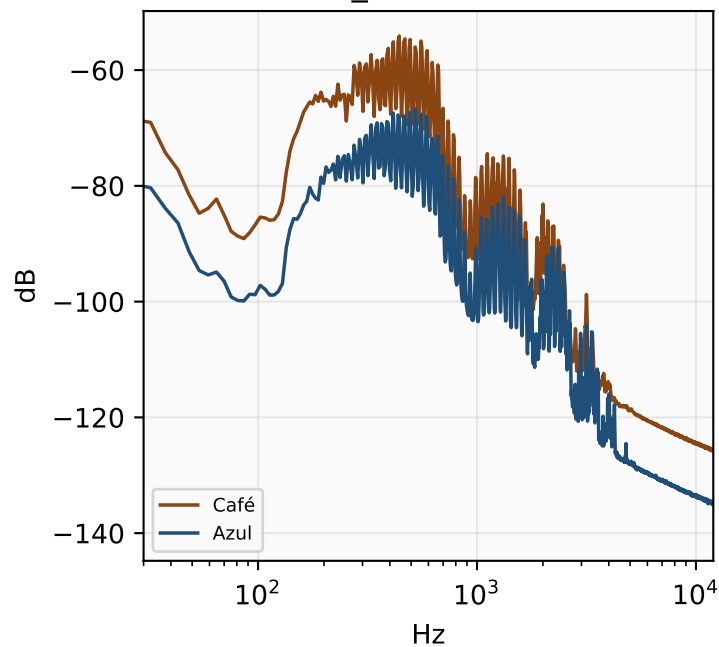
A<sub>12</sub>



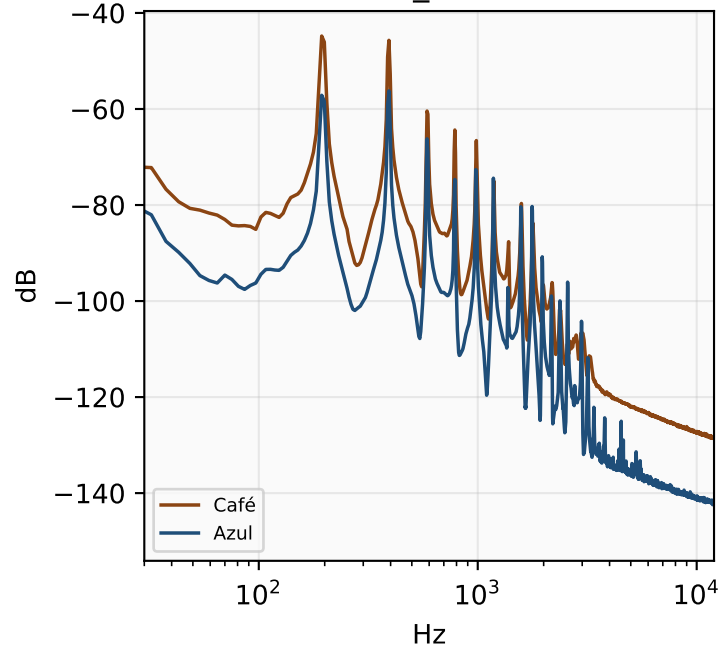
E<sub>12</sub>



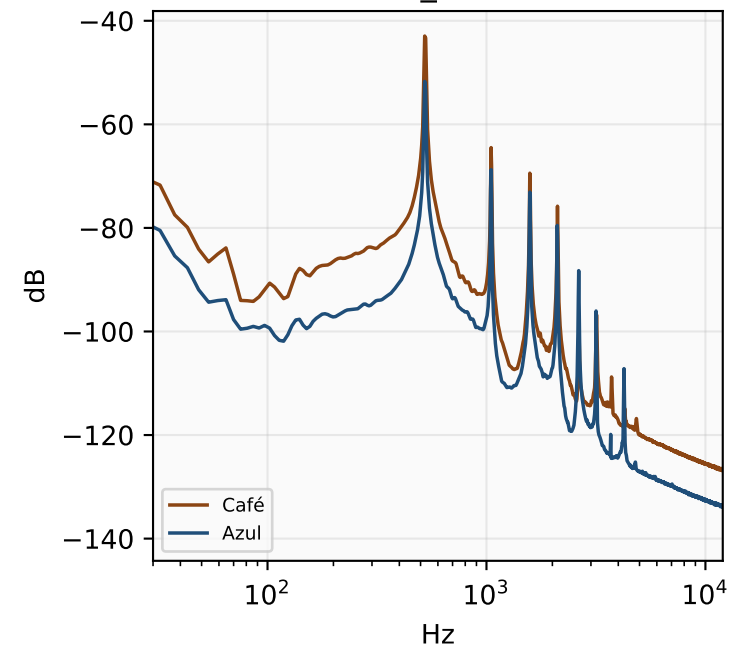
C<sub>chromatic</sub>



G<sub>12</sub>

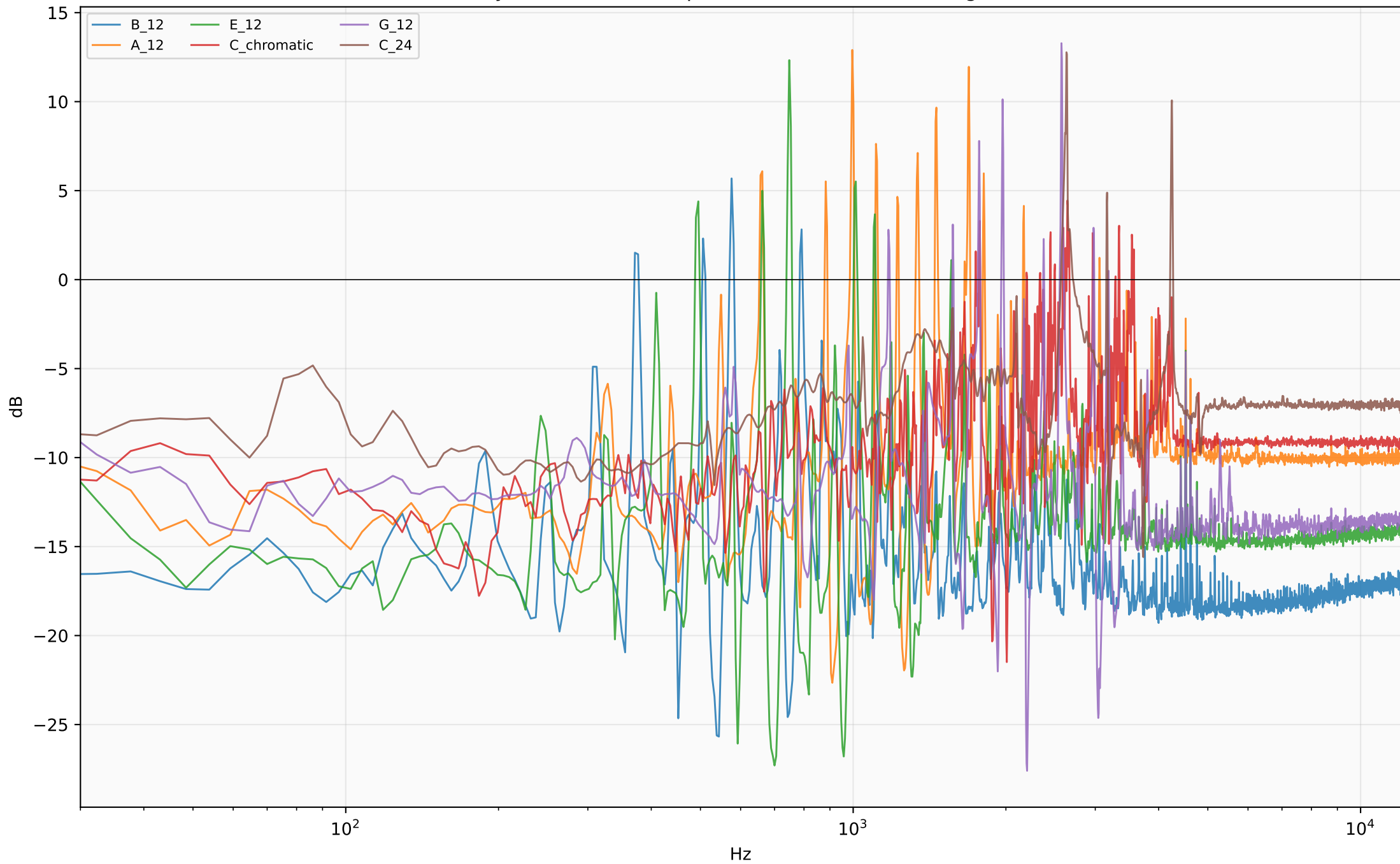


C<sub>24</sub>



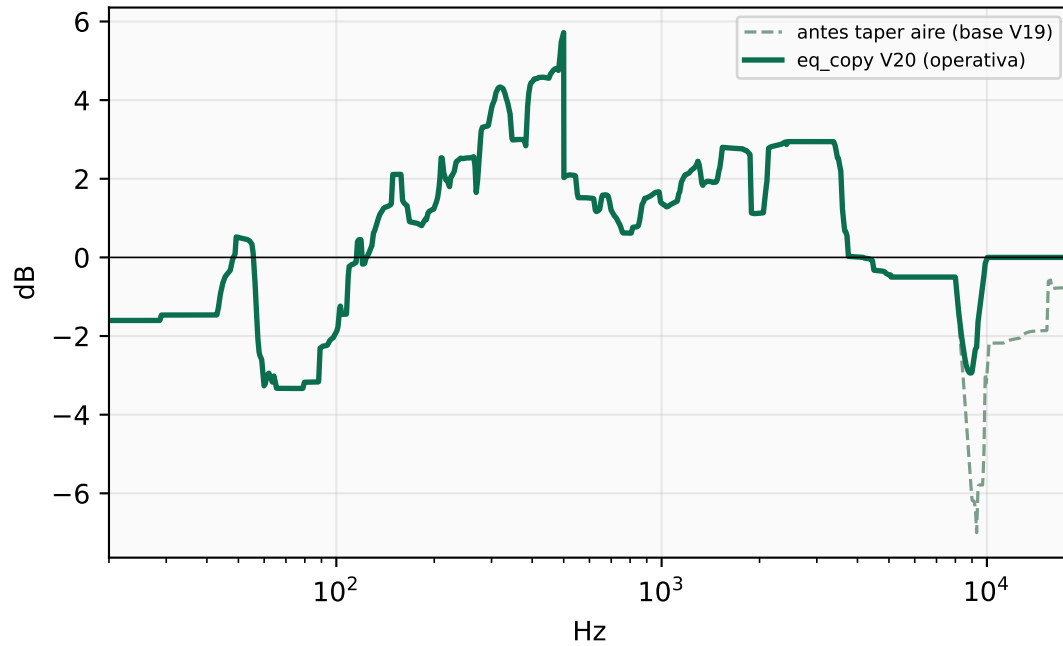
## 1b. Diferencia espectral Azul – Café (por pareja, sin EQ)

Incluye nivel + timbre (por eso  $\sim -12$  dB de offset global)

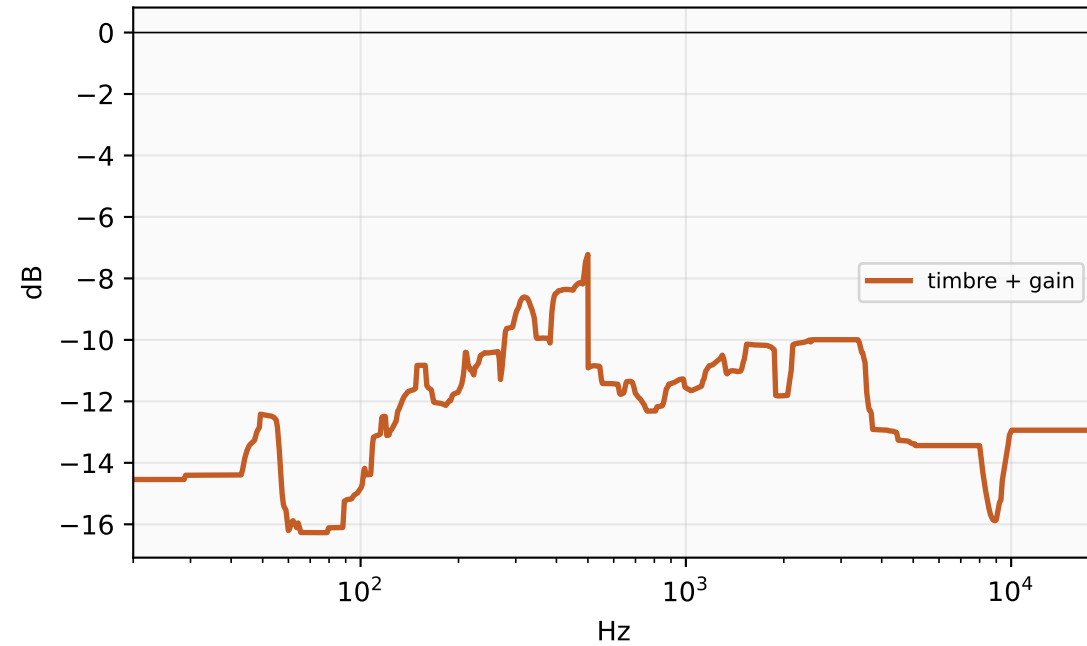


## 2. EQ operativa fiel — V22.0 operativa · air v22\_hard\_10k (gain -12.94 dB)

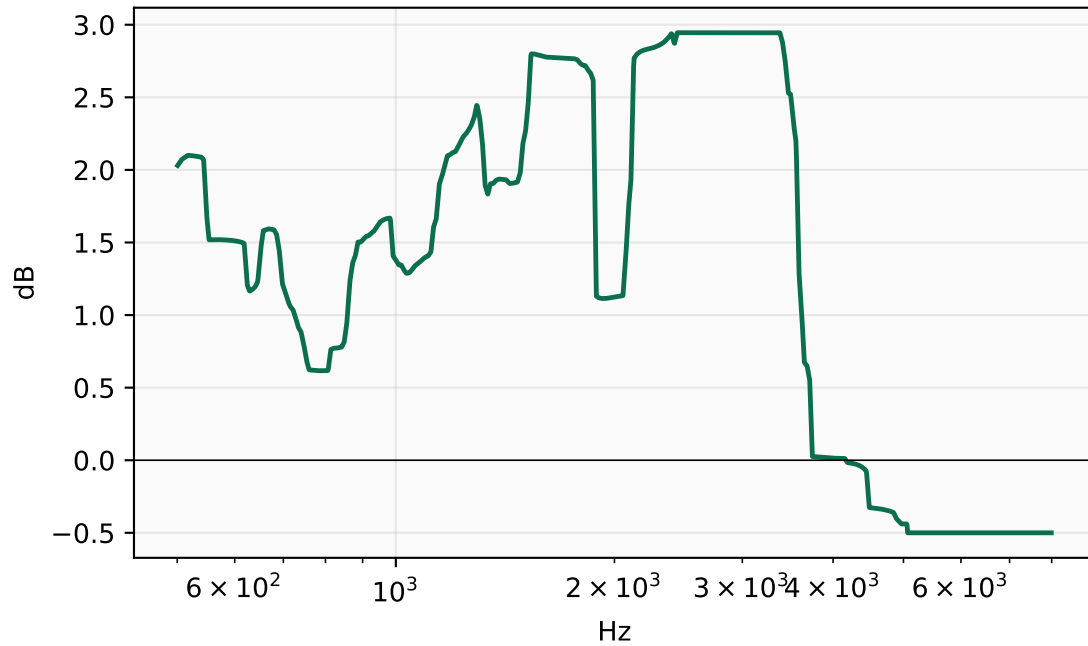
Timbre (sin gain)



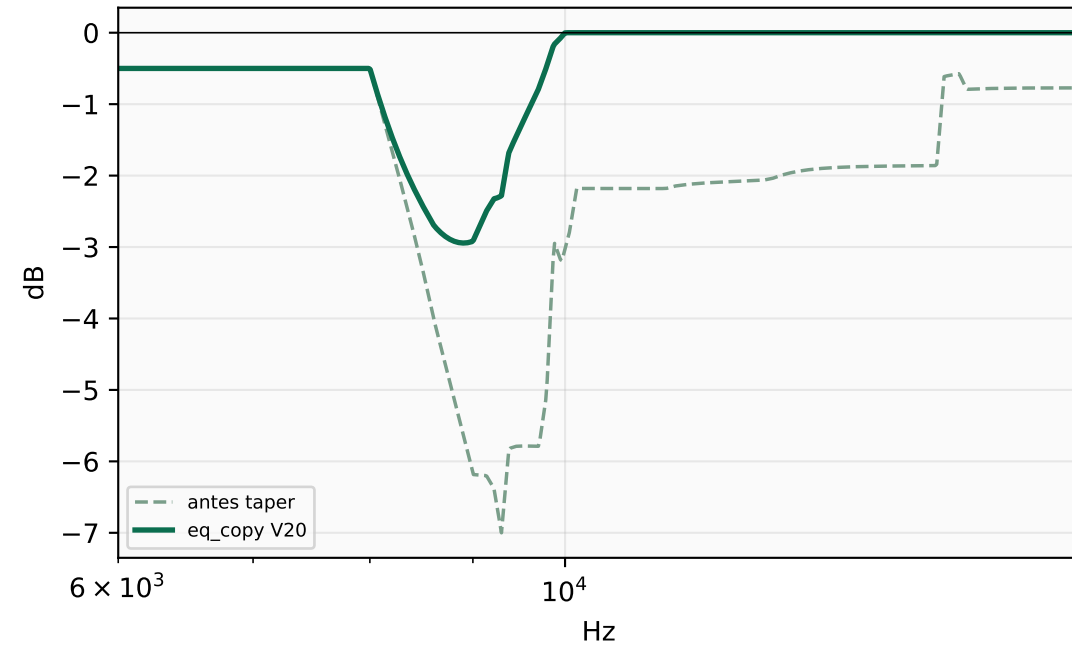
Total a aplicar al Café



Zoom 0.5–8 kHz (copia crítica)

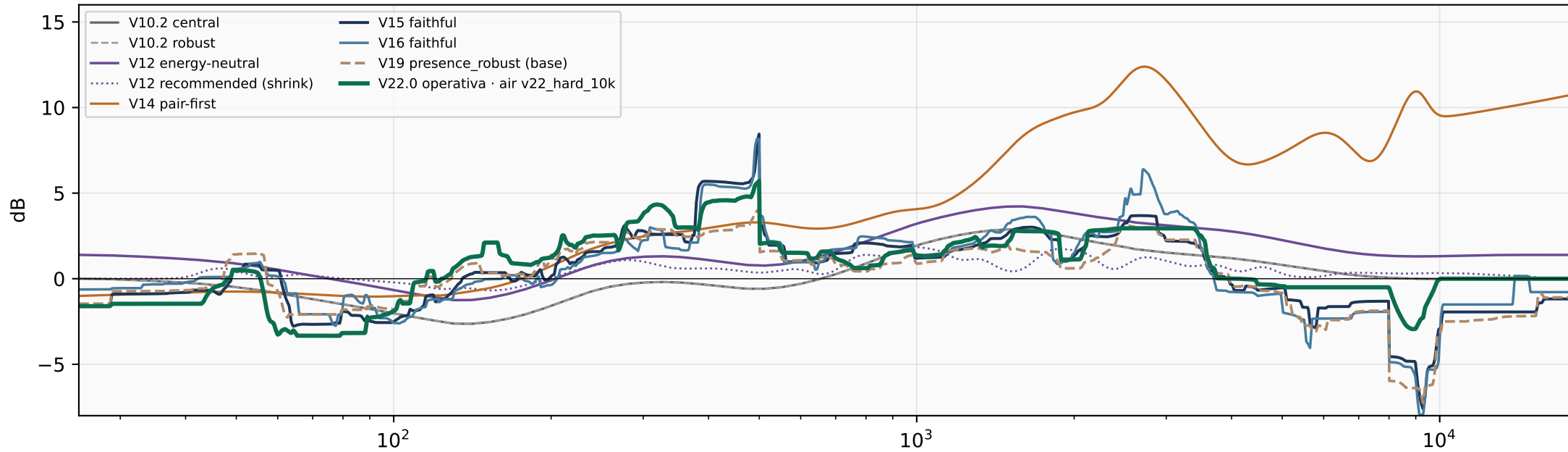


Zoom aire 6–18 kHz (v22\_hard\_10k)

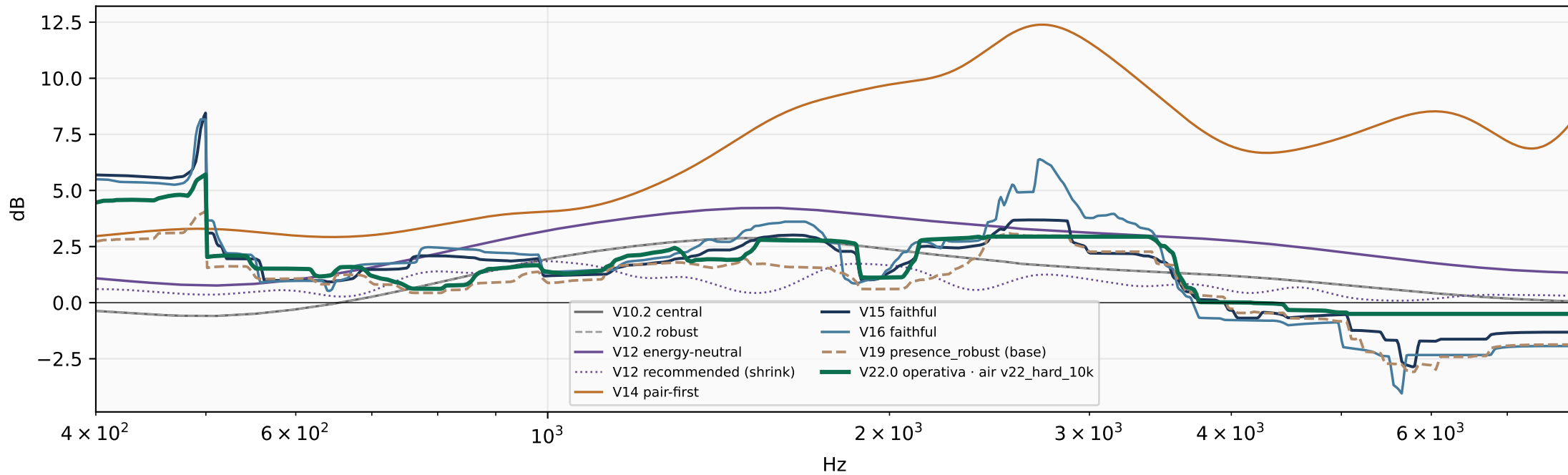


### 3. Todas las curvas de transferencia Café → Azul

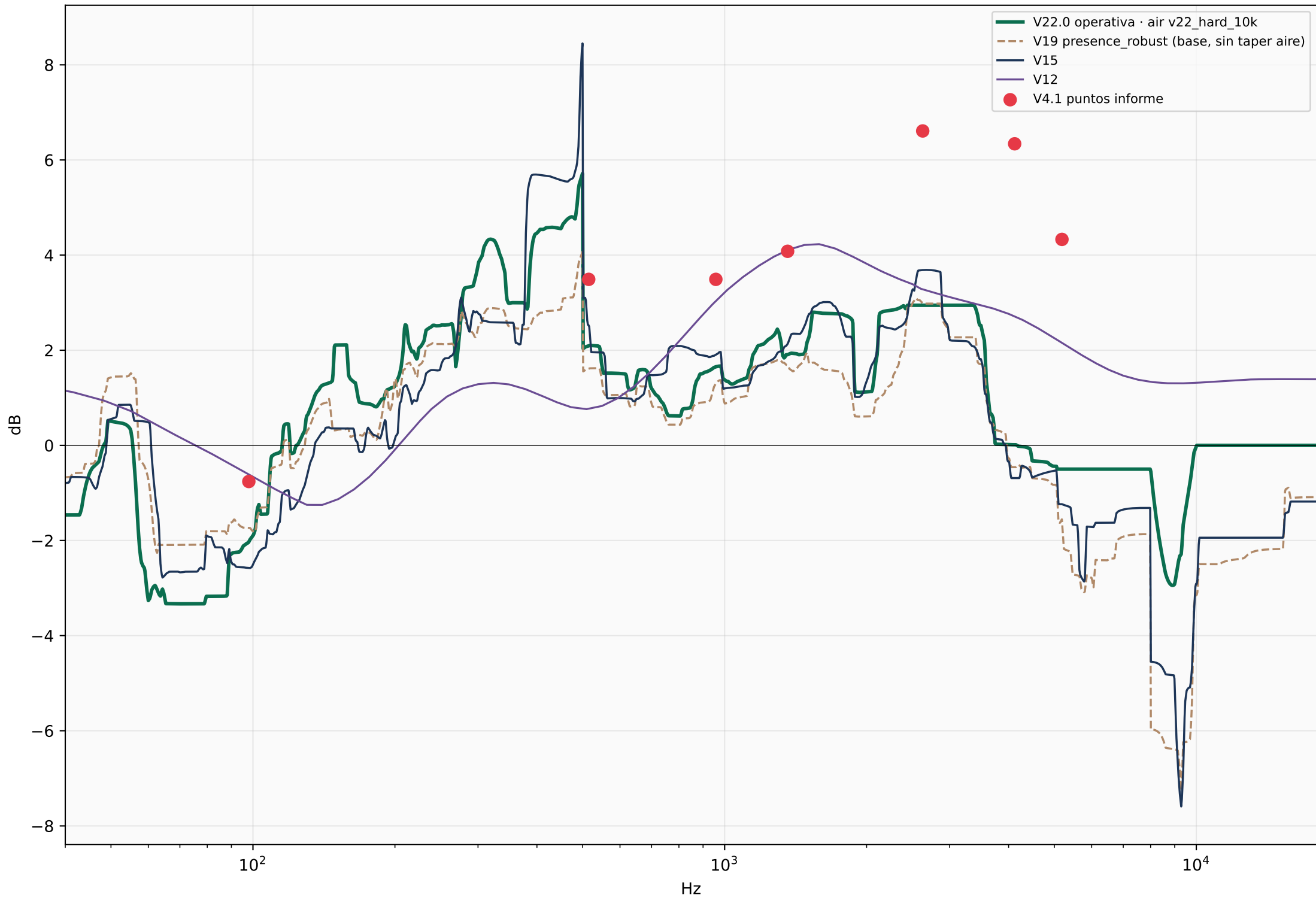
Comparación completa (timbre, sin gain global)



Zoom presencia / medios (0.4-8 kHz)

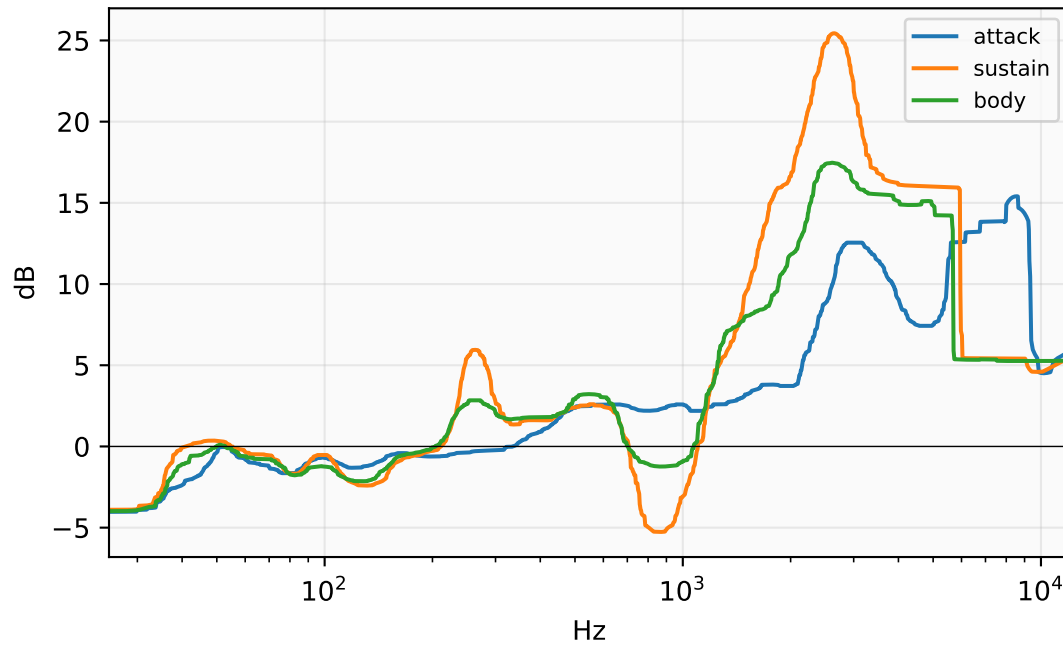


3b. Operativa V20 vs V19 base / puntos V4.1



## 4. Curvas por fase temporal (V13)

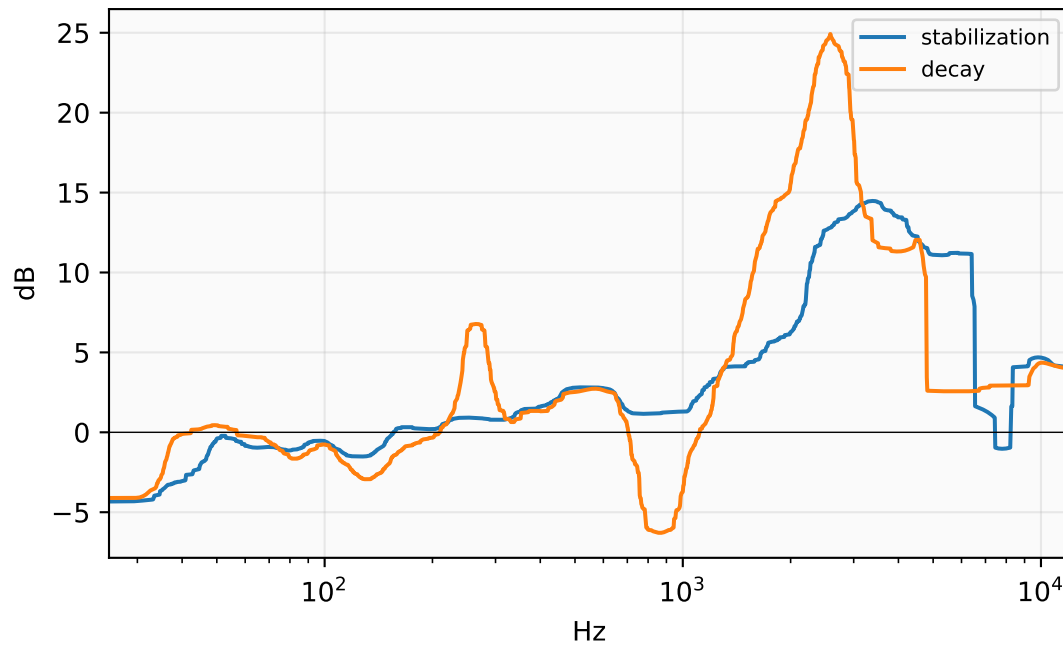
Ataque / sustain / cuerpo



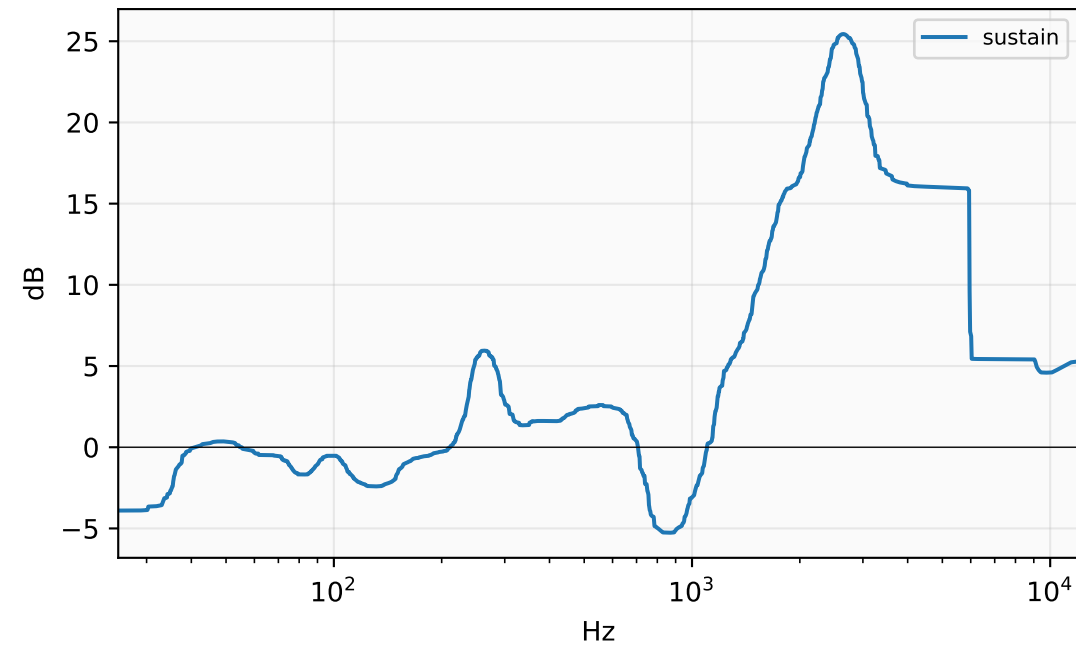
Ataque – cuerpo



Estabilización / decay

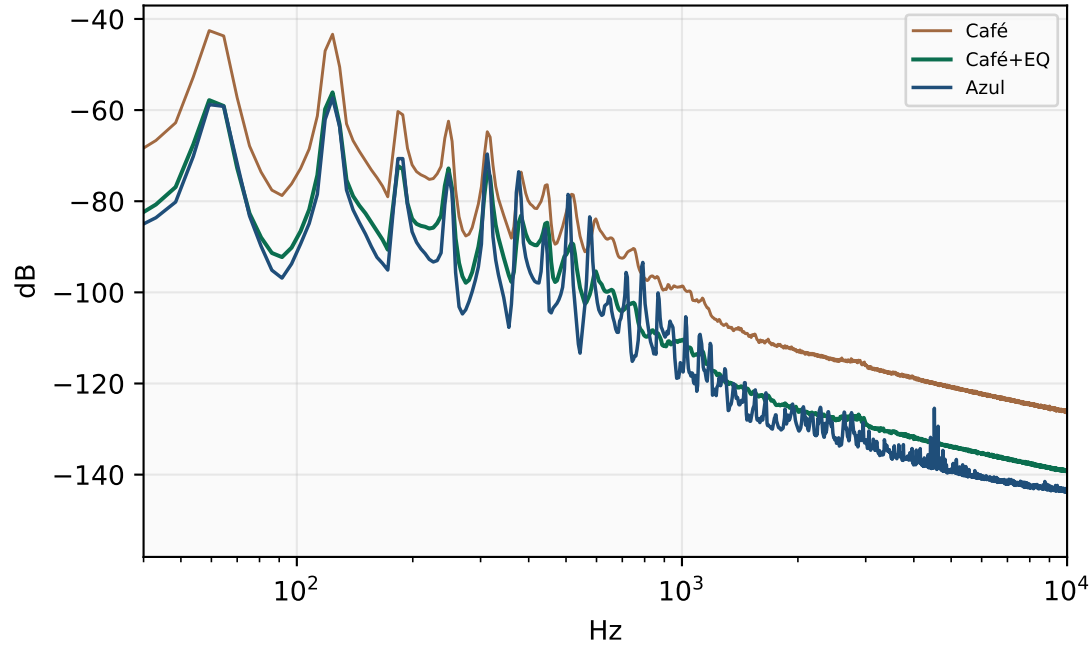


Sustain (detalle)

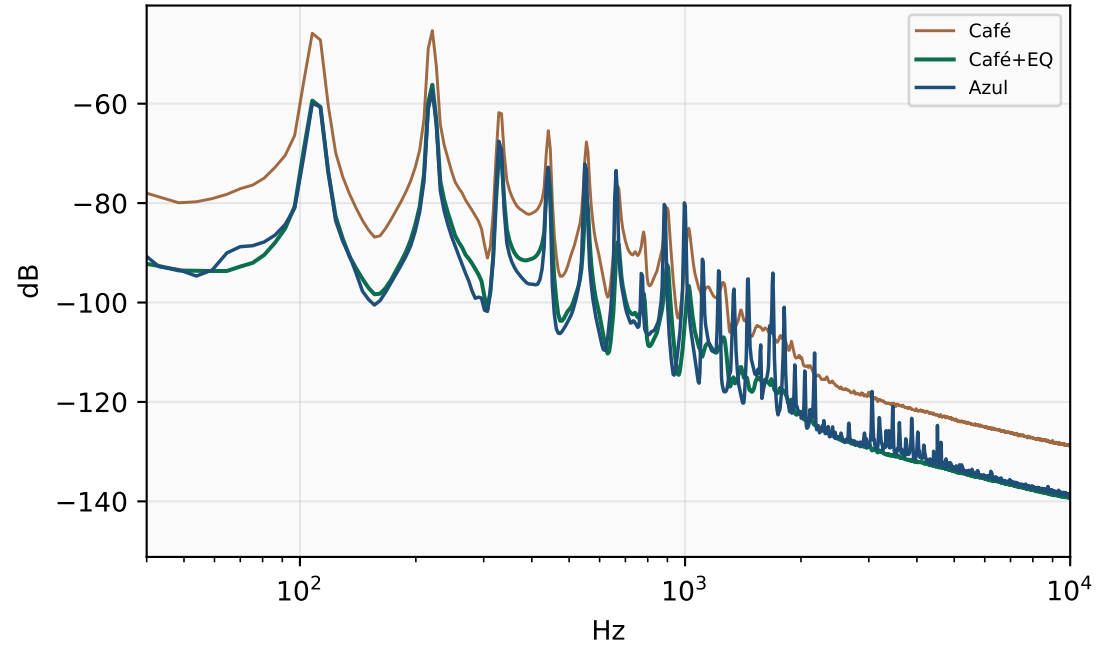


## 5. Fidelidad de copia: Café → (+EQ+gain -12.94 dB) vs Azul

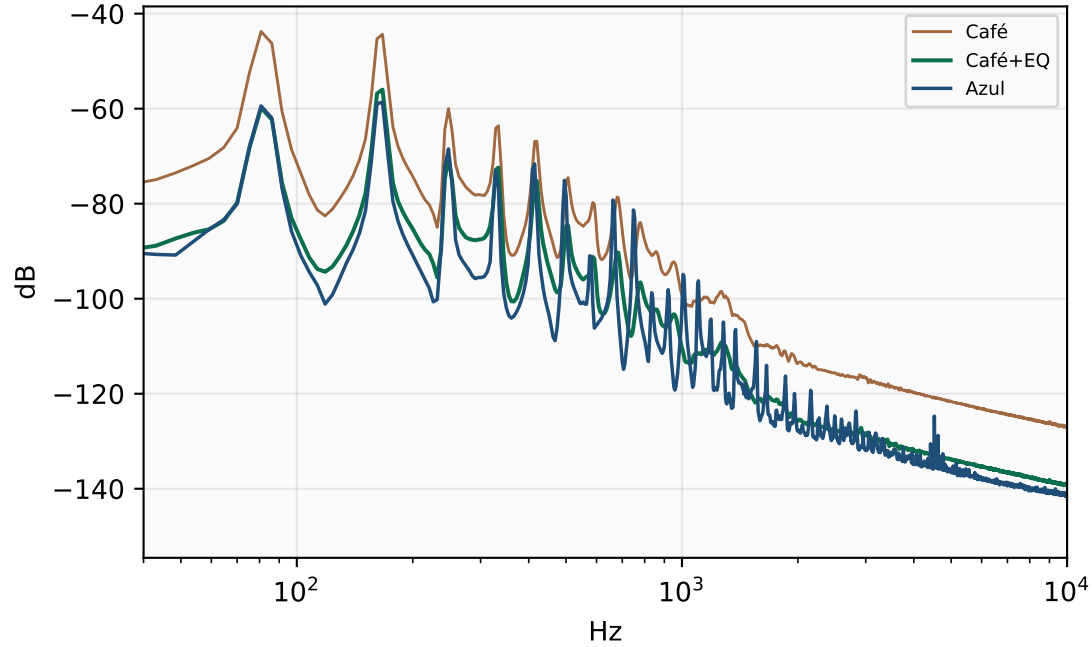
B\_12



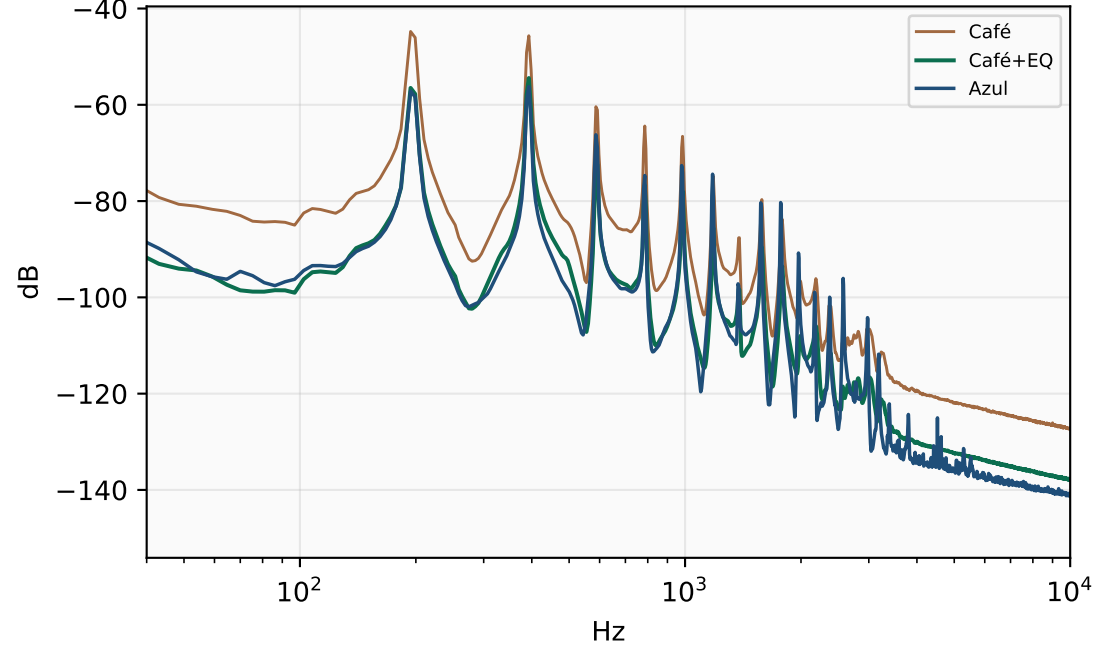
A\_12



E\_12



G\_12



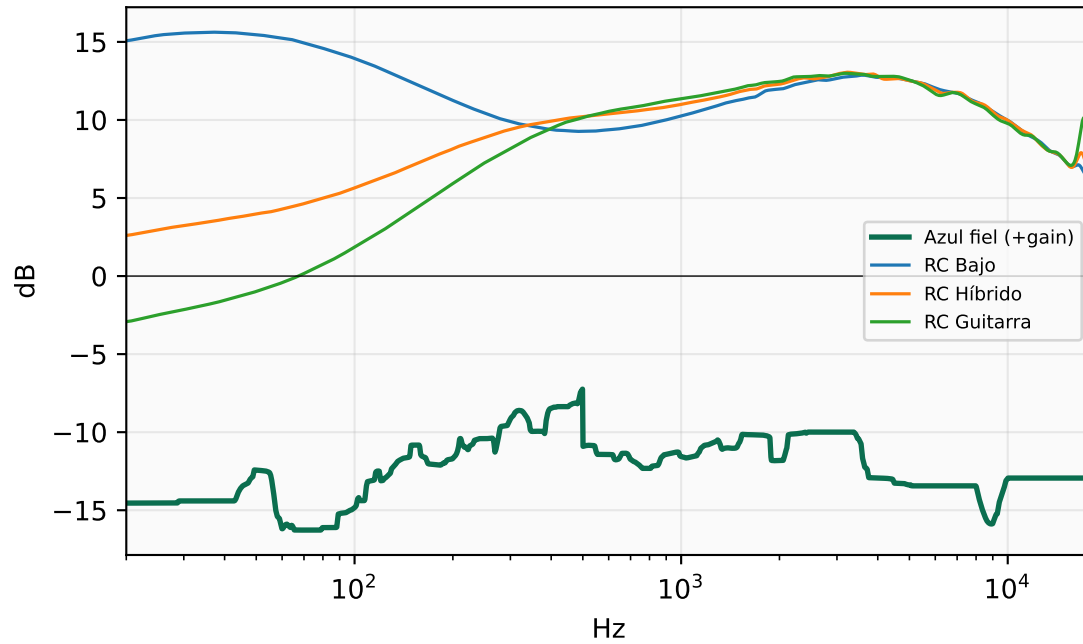


## 5b. Ranking forma agudos V22

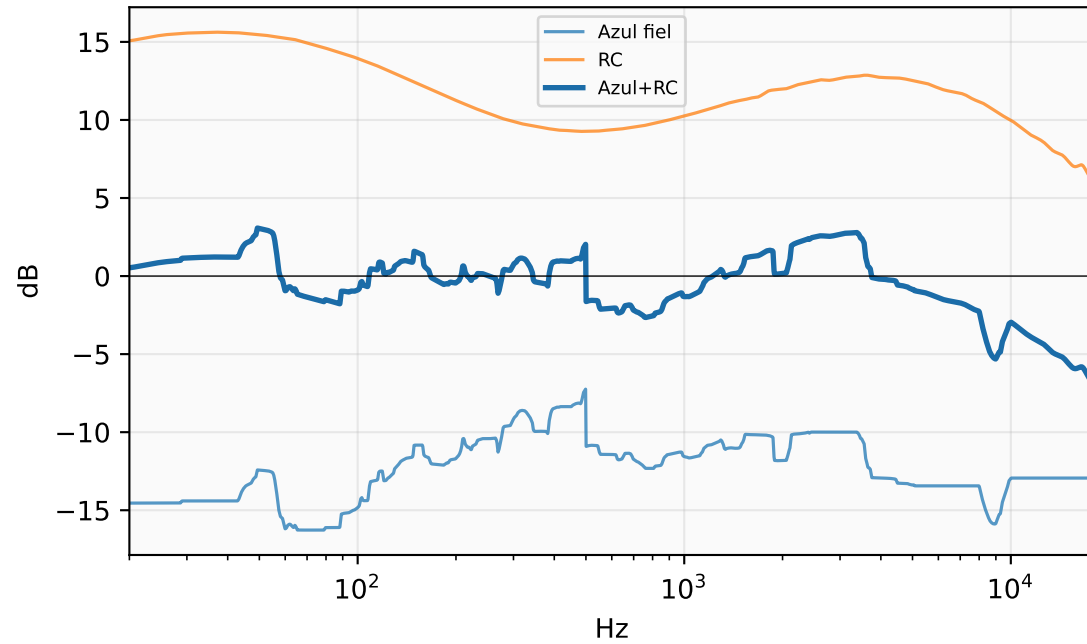
| variant                        | holdout_critical_rmse_db | bias_2k4k_db | gain_db | presence_scale |
|--------------------------------|--------------------------|--------------|---------|----------------|
| v22_winsor_weighted            | 4.578                    | -0.520       | -12.616 | 0.360          |
| v22_winsor_weighted_floor      | 4.570                    | -0.520       | -12.616 | 0.360          |
| v22_winsor_robust              | 4.578                    | -0.520       | -12.616 | 0.360          |
| v22_winsor_robust_floor        | 4.570                    | -0.520       | -12.616 | 0.360          |
| v22_winsor_tight               | 4.591                    | -0.663       | -12.616 | 0.360          |
| v22_winsor_tight_floor         | 4.582                    | -0.663       | -12.616 | 0.360          |
| v22_winsor_fretted_tight       | 4.333                    | -0.507       | -12.939 | 0.300          |
| v22_winsor_fretted_tight_floor | 4.333                    | -0.507       | -12.939 | 0.300          |

## 6. RC + EQ fiel Azul (gain -12.94 dB)

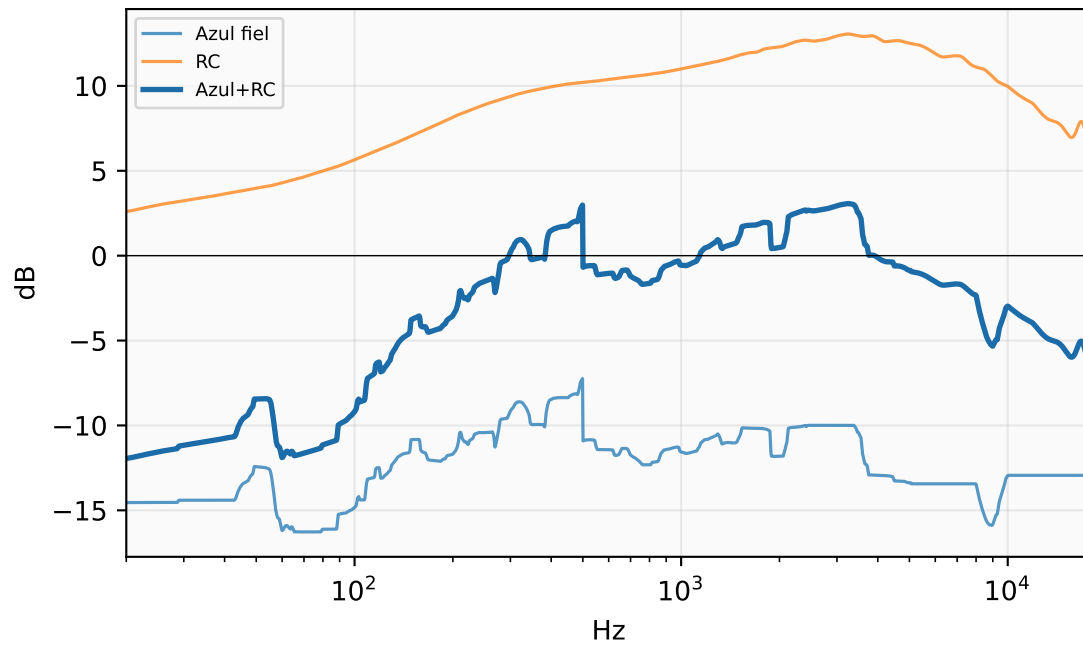
Componentes



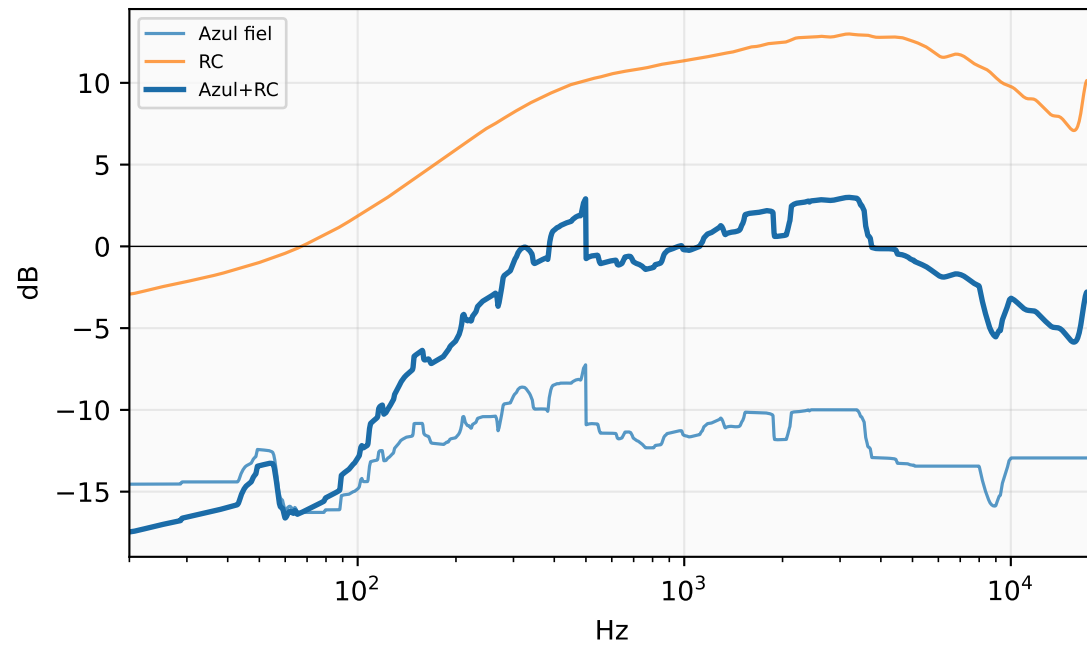
Composición Bajo



Composición Híbrido



Composición Guitarra



# 7. Cierre — estado de la copia Azul

EQ operativa: V22.0 operativa · air v22\_hard\_10k

pipeline\_version: V22.0-operative

source\_variant: v22\_winsor\_fretted\_tight\_floor+v22\_hard\_10k

air\_policy: v22\_hard\_10k

Gain: -12.939 dB (reducir el Café)

Smoothing: none

Puntos de la curva operativa (eq\_copy\_db = V20 con taper aire):

98 Hz → -2.03 dB

515 Hz → +2.10 dB

958 Hz → +1.65 dB

1360 Hz → +1.91 dB

2630 Hz → +2.95 dB

4120 Hz → +0.01 dB

5190 Hz → -0.50 dB

8000 Hz → -0.51 dB

10000 Hz → -0.00 dB

15000 Hz → +0.00 dB

18000 Hz → +0.00 dB

Archivos clave:

· CURVA\_COPIA\_OPERATIVA.csv / GAIN\_COPIA\_OPERATIVA.csv

· RESUMEN\_V20.json / FIDELIDAD\_RANKING\_AIRE\_V20.csv

· FIDELIDAD\_SIN\_SUAVIZAR.md / MAPA\_METODOLOGICO\_V41\_ULTRAPROFUNDO.md

· Audios: renders/FIDELIDAD\_V20/...